

量子场论

第 9 章 分立对称性和 Majorana 旋量场

9.6 节 Weyl、Dirac 和 Majorana 旋量

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9.6 节 Weyl、Dirac 和 Majorana 旋量

9.6.1 小节 左手和右手 Weyl 旋量



Dirac 旋量场和 **Majorana 旋量场**都可以分解为**左手**和**右手**的 Weyl 旋量场



为了更深刻地认识旋量场，本节进一步研究 **Weyl 旋量**



用 $\sigma^\mu = (1, \boldsymbol{\sigma})$ 和 $\bar{\sigma}^\mu = (1, -\boldsymbol{\sigma})$ 定义 **2 × 2 矩阵**

$$s^{\mu\nu} \equiv \frac{i}{4}(\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu)$$



由 $(\sigma^\mu)^\dagger = \sigma^\mu$ 和 $(\bar{\sigma}^\mu)^\dagger = \bar{\sigma}^\mu$ 推出

$$(s^{\mu\nu})^\dagger = -\frac{i}{4}[(\bar{\sigma}^\nu)^\dagger(\sigma^\mu)^\dagger - (\bar{\sigma}^\mu)^\dagger(\sigma^\nu)^\dagger] = -\frac{i}{4}(\bar{\sigma}^\nu \sigma^\mu - \bar{\sigma}^\mu \sigma^\nu) = \frac{i}{4}(\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu)$$



从而将 **Weyl 表象**中的**旋量表示生成元**约化为

$$S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu] = \frac{i}{4} \begin{pmatrix} \sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu & \\ & \bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu \end{pmatrix} = \begin{pmatrix} s^{\mu\nu} & \\ & (s^{\mu\nu})^\dagger \end{pmatrix}$$



也就是说，**4 × 4 矩阵** $S^{\mu\nu}$ 是 **2 × 2 矩阵** $s^{\mu\nu}$ 和 $(s^{\mu\nu})^\dagger$ 的**直和**



因而 $s^{\mu\nu}$ 和 $(s^{\mu\nu})^\dagger$ 是两个 Lorentz 群 **2 维表示**的**生成元**

左手和右手 Weyl 旋量所处 2 维表示

🐱 对于 Lorentz 变换 Λ 的一组变换参数 $\omega_{\mu\nu}$ ，用 $s^{\mu\nu}$ 生成固有保时向有限变换

$$d(\Lambda) \equiv \exp \left(-\frac{i}{2} \omega_{\mu\nu} s^{\mu\nu} \right)$$

👴 它属于左手 Weyl 旋量所处的 2 维表示 $\{d(\Lambda)\}$

👦 相应的逆变换矩阵为 $d^{-1}(\Lambda) = \exp \left(\frac{i}{2} \omega_{\mu\nu} s^{\mu\nu} \right)$ ，取厄米共轭，得

$$d^{-1\dagger}(\Lambda) = \exp \left[-\frac{i}{2} \omega_{\mu\nu} (s^{\mu\nu})^\dagger \right]$$

👧 这是用 $(s^{\mu\nu})^\dagger$ 生成的固有保时向有限变换，属于右手 Weyl 旋量所处的 2 维表示

左手和右手 Weyl 旋量所处 2 维表示

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 于是，旋量表示的 4×4 Lorentz 变换矩阵分解为

$$D(\Lambda) = \exp \left(-\frac{i}{2} \omega_{\mu\nu} S^{\mu\nu} \right) = \begin{pmatrix} e^{-i\omega_{\mu\nu} s^{\mu\nu}/2} & \\ & e^{-i\omega_{\mu\nu} (s^{\mu\nu})^\dagger/2} \end{pmatrix} = \begin{pmatrix} d(\Lambda) & \\ & d^{-1\dagger}(\Lambda) \end{pmatrix}$$

 因此，4 维旋量表示 $\{D(\Lambda)\}$ 是 2 维表示 $\{d(\Lambda)\}$ 和 $\{d^{-1\dagger}(\Lambda)\}$ 的直和

等价表示

 利用 $\sigma^2 \sigma^\mu \sigma^2 = (\bar{\sigma}^\mu)^T$ 和 $\sigma^2 \bar{\sigma}^\mu \sigma^2 = (\sigma^\mu)^T$ 推出

$$\begin{aligned}\sigma^2 s^{\mu\nu} \sigma^2 &= \frac{i}{4} (\sigma^2 \sigma^\mu \sigma^2 \sigma^2 \bar{\sigma}^\nu \sigma^2 - \sigma^2 \sigma^\nu \sigma^2 \sigma^2 \bar{\sigma}^\mu \sigma^2) \\ &= \frac{i}{4} [(\bar{\sigma}^\mu)^T (\sigma^\nu)^T - (\bar{\sigma}^\nu)^T (\sigma^\mu)^T] = -(s^{\mu\nu})^T\end{aligned}$$

$$\begin{aligned}\sigma^2 d(\Lambda) \sigma^2 &= \exp \left(-\frac{i}{2} \omega_{\mu\nu} \sigma^2 s^{\mu\nu} \sigma^2 \right) \\ &= \exp \left[\frac{i}{2} \omega_{\mu\nu} (s^{\mu\nu})^T \right] = \left[\exp \left(\frac{i}{2} \omega_{\mu\nu} s^{\mu\nu} \right) \right]^T = d^{-1T}(\Lambda)\end{aligned}$$

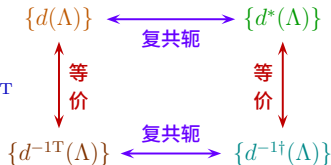
 这里 $d^{-1T}(\Lambda) = [d^{-1\dagger}(\Lambda)]^*$ ，线性表示 $\{d^{-1T}(\Lambda)\}$ 是 $\{d^{-1\dagger}(\Lambda)\}$ 的复共轭表示

等价表示

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$$\begin{aligned}\sigma^2 d(\Lambda) \sigma^2 &= \exp \left(-\frac{i}{2} \omega_{\mu\nu} \sigma^2 s^{\mu\nu} \sigma^2 \right) \\ &= \exp \left[\frac{i}{2} \omega_{\mu\nu} (s^{\mu\nu})^T \right] = \left[\exp \left(\frac{i}{2} \omega_{\mu\nu} s^{\mu\nu} \right) \right]^T = d^{-1T}(\Lambda)\end{aligned}$$



🥕 这里 $d^{-1T}(\Lambda) = [d^{-1†}(\Lambda)]^*$ ，线性表示 $\{d^{-1T}(\Lambda)\}$ 是 $\{d^{-1†}(\Lambda)\}$ 的复共轭表示

🧑 将 Pauli 矩阵 σ^2 看作一个么正变换矩阵，满足 $(\sigma^2)^{-1} = (\sigma^2)^\dagger = \sigma^2$

👨 则 $d(\Lambda)$ 与 $d^{-1T}(\Lambda)$ 由一个相似变换联系起来，相似变换矩阵为 σ^2

👨 根据 1.4 节定义，线性表示 $\{d(\Lambda)\}$ 和 $\{d^{-1T}(\Lambda)\}$ 是等价的

👨 由于 $(\sigma^2)^* = -\sigma^2$ ， $\sigma^2 d(\Lambda) \sigma^2 = d^{-1T}(\Lambda)$ 的复共轭为 $\sigma^2 d^*(\Lambda) \sigma^2 = d^{-1†}(\Lambda)$

👨 可见，线性表示 $\{d(\Lambda)\}$ 的复共轭表示 $\{d^*(\Lambda)\}$ 与 $\{d^{-1†}(\Lambda)\}$ 等价

左手 Weyl 旋量

🐎 于是，左手 Weyl 旋量

$$\eta_a = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

的固有保时向 Lorentz 变换为

$$\eta'_a = [d(\Lambda)]_a{}^b \eta_b, \quad a, b = 1, 2$$

👤 η_a 是 $\{d(\Lambda)\}$ 表示空间中的列矢量

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 η_a 是 $\{d(\Lambda)\}$ 表示空间中的列矢量

 引入反对称的二维 Levi-Civita 符号 ϵ^{ab} ，定义为

$$\epsilon^{12} = -\epsilon^{21} = 1, \quad \epsilon^{11} = \epsilon^{22} = 0$$

 它与 Pauli 矩阵 σ^2 的关系是

$$\epsilon^{ab} = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} = i \begin{pmatrix} & -i \\ i & \end{pmatrix} = (i\sigma^2)^{ab}$$

等价的左手 Weyl 旋量

🐕 通过 ε^{ab} 定义

$$\eta^a \equiv \varepsilon^{ab} \eta_b = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} \eta_2 \\ -\eta_1 \end{pmatrix}$$

👮 则

$$\eta^1 = \eta_2, \quad \eta^2 = -\eta_1$$

👩 $\sigma^2 d(\Lambda) \sigma^2 = d^{-1T}(\Lambda)$ 等价于 $\sigma^2 d(\Lambda) = d^{-1T}(\Lambda) \sigma^2$ ，故 η^a 的 Lorentz 变换为

$$\begin{aligned} \eta'^a &= \varepsilon^{ab} \eta'_b = \varepsilon^{ab} [d(\Lambda)]_b^c \eta_c = i[\sigma^2 d(\Lambda)]^{ac} \eta_c \\ &= i[d^{-1T}(\Lambda) \sigma^2]^{ac} \eta_c = [d^{-1T}(\Lambda)]^a_b \varepsilon^{bc} \eta_c \end{aligned}$$

等价的左手 Weyl 旋量

 通过 ε^{ab} 定义

$$\eta^a \equiv \varepsilon^{ab} \eta_b = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} \eta_2 \\ -\eta_1 \end{pmatrix}$$

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 即

$$\eta'^a = [d^{-1T}(\Lambda)]^a_b \eta^b$$

 可见 η^a 是 $\{d^{-1T}(\Lambda)\}$ 表示空间中的列矢量 由于 $\{d^{-1T}(\Lambda)\}$ 等价于 $\{d(\Lambda)\}$ ， η^a 也是左手 Weyl 旋量

ϵ^{ab} 和 ϵ_{ab}

🐶 两种左手 Weyl 旋量 η_a 与 η^a 是等价的，它们之间的关系类似于 Lorentz 逆变矢量 A^μ 与协变矢量 $A_\mu = g_{\mu\nu} A^\nu$ 之间的关系

🧑 ϵ^{ab} 的作用类似于度规 $g_{\mu\nu}$ ，相当于 2 维旋量空间的“度规”，用于提升旋量指标

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👷 用 $\epsilon_{12} = -\epsilon_{21} = -1$ 和 $\epsilon_{11} = \epsilon_{22} = 0$ 定义 ϵ_{ab} ，则

$$\epsilon_{ab} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix} = -i \begin{pmatrix} & -i \\ i & \end{pmatrix} = (-i\sigma^2)_{ab}$$

👷 ϵ_{ab} 是 ϵ^{ab} 的逆矩阵，满足

$$\epsilon_{ab}\epsilon^{bc} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix} \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} = \delta_a^c$$

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👤 于是， $\eta^1 = \eta_2$ 和 $\eta^2 = -\eta_1$ 表明

$$\eta_a = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} -\eta^2 \\ \eta^1 \end{pmatrix} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix} \begin{pmatrix} \eta^1 \\ \eta^2 \end{pmatrix} = \epsilon_{ab}\eta^b$$

👤 也就是说， ϵ_{ab} 用于下降旋量指标

左手 Weyl 旋量的内积

🐷 任意两个左手 Weyl 旋量 η_a 和 ζ_a 的内积

$$\eta^a \zeta_a = \varepsilon^{ab} \eta_b \zeta_a = \varepsilon_{ab} \eta^a \zeta^b$$

在固有保时向 Lorentz 变换下不变，满足

$$\eta'^a \zeta'_a = [d^{-1T}(\Lambda)]^a_b \eta^b [d(\Lambda)]_a^c \zeta_c = \eta^b [d^{-1}(\Lambda)]_b^a [d(\Lambda)]_a^c \zeta_c = \eta^b \delta_b^c \zeta_c = \eta^a \zeta_a$$



第二步用了转置性质 $[d^{-1T}(\Lambda)]^a_b = [d^{-1}(\Lambda)]_b^a$ ，可见 $\eta^a \zeta_a$ 是 Lorentz 标量

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$$\eta'^a \zeta'_a = [d^{-1T}(\Lambda)]^a_b \eta^b [d(\Lambda)]_a^c \zeta_c = \eta^b [d^{-1}(\Lambda)]_b^a [d(\Lambda)]_a^c \zeta_c = \eta^b \delta_b^c \zeta_c = \eta^a \zeta_a$$

👩 第二步用了**转置**性质 $[d^{-1T}(\Lambda)]^a_b = [d^{-1}(\Lambda)]_b^a$ ，可见 $\eta^a \zeta_a$ 是 **Lorentz 标量**

👧 由 $\eta^1 = \eta_2$ 、 $\eta^2 = -\eta_1$ 、 $\zeta^1 = \zeta_2$ 和 $\zeta^2 = -\zeta_1$ 得

$$\eta^a \zeta_a = \eta^1 \zeta_1 + \eta^2 \zeta_2 = \eta_2 \zeta_1 - \eta_1 \zeta_2 = -\eta_2 \zeta^2 - \eta_1 \zeta^1 = -\eta_a \zeta^a$$

👧 即参与缩并的**旋量指标一升一降**会多出一个**负号**

👧 这种性质与 Lorentz 矢量内积 $A^\mu B_\mu = A_\mu B^\mu$ **截然不同**

👧 原因在于旋量空间“度规” ε^{ab} 是**反对称**的

Grassmann 数

🐑 $\eta^a \zeta_a = -\eta_a \zeta^a$ 表明 $\eta^a \eta_a = -\eta_a \eta^a$ ，若 η_a 和 η^a 是普通的复数，则 $\eta^a \eta_a = 0$

🦸 为了使 $\eta^a \eta_a \neq 0$ ，必须要求左手 Weyl 旋量 η^a 与 η_a 反对易

🧐 即它们是 Grassmann 数，任意两个 Grassmann 数都是反对易的

🧙 以复数作为组合系数，则若干个 Grassmann 数的线性组合也是 Grassmann 数

🧚 因此， η_a 是 Grassmann 数意味着 $\eta^a = \varepsilon^{ab} \eta_b$ 也是 Grassmann 数

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🦊 虽然如此，Grassmann 数是反对易的 c 数，而不是 Hilbert 空间上的算符

🐉 对 Grassmann 数表达的旋量场进行量子化，才得到旋量场算符，而 Grassmann 数的反对易性质与旋量场算符的反对易关系相匹配

⚠️ 旋量也可以不是 Grassmann 数，旋量系数 $u(\mathbf{p}, \lambda)$ 和 $v(\mathbf{p}, \lambda)$ 就是普通的复数

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🧚 假设 η_a 和 ζ^a 都是 Grassmann 数，则 $\eta_a \zeta^a = -\zeta^a \eta_a$ ，相应地，将省略旋量指标的内积写成 $\eta \zeta \equiv \eta^a \zeta_a = -\eta_a \zeta^a = \zeta^a \eta_a = \zeta \eta$ ，即内积 $\eta \zeta$ 和 $\zeta \eta$ 是相等的

🧚 内积 $\eta^a \eta_a$ 有等价表达式
$$\begin{aligned} \eta \eta &= \eta^a \eta_a = \varepsilon_{ab} \eta^a \eta^b = -\eta^1 \eta^2 + \eta^2 \eta^1 = -2\eta^1 \eta^2 \\ &= 2\eta_2 \eta_1 = \eta_2 \eta_1 - \eta_1 \eta_2 = -\varepsilon^{ab} \eta_a \eta_b = -\eta_a \eta^a \end{aligned}$$

左手 Weyl 旋量的复共轭

🐑 将左手 Weyl 旋量 η_a 的复共轭记为 $\eta_a^\dagger = \begin{pmatrix} \eta_1^\dagger \\ \eta_2^\dagger \end{pmatrix}$

🧢 量子化之后，算符 η_a 和 $\eta_a^\dagger$ 互为厄米共轭

🕶️ 对 $\eta'_a = [d(\Lambda)]_a{}^b \eta_b$ 两边取复共轭，得到 $\eta_a^\dagger$ 的 Lorentz 变换

$$\eta_a'^\dagger = [d^*(\Lambda)]_{\dot{a}}{}^{\dot{b}} \eta_{\dot{b}}^\dagger$$

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$$\eta_a'^\dagger = [d^*(\Lambda)]_{\dot{a}}^{\dot{b}} \eta_{\dot{b}}^\dagger$$

👕 引进指标上带着点号的二维 Levi-Civita 符号

$$\epsilon^{\dot{a}\dot{b}} = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} = (i\sigma^2)^{\dot{a}\dot{b}}, \quad \epsilon_{\dot{a}\dot{b}} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix} = (-i\sigma^2)_{\dot{a}\dot{b}}$$

👖 其分量数值与 ϵ^{ab} 和 ϵ_{ab} 分别相同

👣 定义 $\eta^{\dot{a}} \equiv \epsilon^{\dot{a}\dot{b}} \eta_{\dot{b}}^\dagger = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \begin{pmatrix} \eta_1^\dagger \\ \eta_2^\dagger \end{pmatrix} = \begin{pmatrix} \eta_2^\dagger \\ -\eta_1^\dagger \end{pmatrix}$, 则有 $\eta^{\dot{1}} = \eta_2^\dagger$ 和 $\eta^{\dot{2}} = -\eta_1^\dagger$

右手 Weyl 旋量

🐐 $\sigma^2 d^*(\Lambda) \sigma^2 = d^{-1\dagger}(\Lambda)$ 等价于 $\sigma^2 d^*(\Lambda) = d^{-1\dagger}(\Lambda) \sigma^2$

🧑‍🎓 故 $\eta^{\dagger\dot{a}}$ 的 Lorentz 变换为

$$\begin{aligned}\eta'^{\dagger\dot{a}} &= \varepsilon^{\dot{a}\dot{b}} \eta_b^\dagger = \varepsilon^{\dot{a}\dot{b}} [d^*(\Lambda)]_{\dot{b}}^{\dot{c}} \eta_c^\dagger = i[\sigma^2 d^*(\Lambda)]^{\dot{a}\dot{c}} \eta_c^\dagger \\ &= i[d^{-1\dagger}(\Lambda) \sigma^2]^{\dot{a}\dot{c}} \eta_c^\dagger = [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{b}} \varepsilon^{\dot{b}\dot{c}} \eta_c^\dagger\end{aligned}$$

🎀 即

$$\eta'^{\dagger\dot{a}} = [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{b}} \eta^{\dagger\dot{b}}$$

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🐐 $\sigma^2 d^*(\Lambda) \sigma^2 = d^{-1\dagger}(\Lambda)$ 等价于 $\sigma^2 d^*(\Lambda) = d^{-1\dagger}(\Lambda) \sigma^2$

🧺 故 $\eta^{\dagger\dot{a}}$ 的 Lorentz 变换为

$$\begin{aligned}\eta'^{\dagger\dot{a}} &= \varepsilon^{\dot{a}\dot{b}} \eta'_b{}^\dagger = \varepsilon^{\dot{a}\dot{b}} [d^*(\Lambda)]_{\dot{b}}{}^{\dot{c}} \eta_c^\dagger = i[\sigma^2 d^*(\Lambda)]^{\dot{a}\dot{c}} \eta_c^\dagger \\ &= i[d^{-1\dagger}(\Lambda) \sigma^2]^{\dot{a}\dot{c}} \eta_c^\dagger = [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{b}} \varepsilon^{\dot{b}\dot{c}} \eta_c^\dagger\end{aligned}$$

🎀 即

$$\eta'^{\dagger\dot{a}} = [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{b}} \eta^{\dagger\dot{b}}$$

👗 可见, $\eta^{\dagger\dot{a}}$ 是 $\{d^{-1\dagger}(\Lambda)\}$ 表示空间中的列矢量, 因而是右手 Weyl 旋量

💍 由于表示 $\{d^*(\Lambda)\}$ 等价于 $\{d^{-1\dagger}(\Lambda)\}$, $\eta_a^\dagger$ 也是右手 Weyl 旋量

💊 因此, 在这套符号约定中, 不带点的旋量指标对应于左手 Weyl 旋量及其表示

👠 而带点的旋量指标对应于右手 Weyl 旋量及其表示

右手 Weyl 旋量的内积

 任意两个右手 Weyl 旋量 $\eta^{\dagger\dot{a}}$ 和 $\zeta^{\dagger\dot{a}}$ 的内积

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = \varepsilon_{\dot{a}\dot{b}} \eta^{\dagger\dot{b}} \zeta^{\dagger\dot{a}} = \varepsilon^{\dot{a}\dot{b}} \eta_a^\dagger \zeta_b^\dagger$$

在固有保时向 Lorentz 变换下**不变**，满足

$$\eta_a^{\prime\dagger} \zeta^{\prime\dagger\dot{a}} = [d^*(\Lambda)]_{\dot{a}}^{\dot{b}} \eta_b^\dagger [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_b^\dagger [d^\dagger(\Lambda)]^{\dot{b}}_{\dot{a}} [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_b^\dagger \delta^{\dot{b}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_a^\dagger \zeta^{\dagger\dot{a}}$$

 第二步用了**转置**性质 $[d^*(\Lambda)]_{\dot{a}}^{\dot{b}} = [d^\dagger(\Lambda)]^{\dot{b}}_{\dot{a}}$ ，可见 $\eta_a^\dagger \zeta^{\dagger\dot{a}}$ 是 **Lorentz 标量**

右手 Weyl 旋量的内积

 任意两个右手 Weyl 旋量 $\eta^{\dagger\dot{a}}$ 和 $\zeta^{\dagger\dot{a}}$ 的内积

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = \varepsilon_{\dot{a}\dot{b}} \eta^{\dagger\dot{b}} \zeta^{\dagger\dot{a}} = \varepsilon^{\dot{a}\dot{b}} \eta_a^\dagger \zeta_b^\dagger$$

在固有保时向 Lorentz 变换下**不变**，满足

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = [d^*(\Lambda)]_{\dot{a}}^{\dot{b}} \eta_b^\dagger [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_b^\dagger [d^\dagger(\Lambda)]^{\dot{b}}_{\dot{a}} [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_b^\dagger \delta^{\dot{b}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_a^\dagger \zeta^{\dagger\dot{a}}$$

 第二步用了**转置**性质 $[d^*(\Lambda)]_{\dot{a}}^{\dot{b}} = [d^\dagger(\Lambda)]^{\dot{b}}_{\dot{a}}$ ，可见 $\eta_a^\dagger \zeta^{\dagger\dot{a}}$ 是 **Lorentz 标量** 由 $\eta^{\dagger\dot{1}} = \eta_2^\dagger$ 、 $\eta^{\dagger\dot{2}} = -\eta_1^\dagger$ 、 $\zeta^{\dagger\dot{1}} = \zeta_2^\dagger$ 和 $\zeta^{\dagger\dot{2}} = -\zeta_1^\dagger$ 得

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = \eta_1^\dagger \zeta^{\dagger\dot{1}} + \eta_2^\dagger \zeta^{\dagger\dot{2}} = -\eta^{\dagger\dot{2}} \zeta^{\dagger\dot{1}} + \eta^{\dagger\dot{1}} \zeta^{\dagger\dot{2}} = -\eta^{\dagger\dot{2}} \zeta_2^\dagger - \eta^{\dagger\dot{1}} \zeta_1^\dagger = -\eta^{\dagger\dot{a}} \zeta_a^\dagger$$

 即参与缩并的**带点旋量指标一升一降**会多出一个**负号**

右手 Weyl 旋量的内积

 任意两个右手 Weyl 旋量 $\eta^{\dagger\dot{a}}$ 和 $\zeta^{\dagger\dot{a}}$ 的内积

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = \varepsilon_{\dot{a}\dot{b}} \eta^{\dagger\dot{b}} \zeta^{\dagger\dot{a}} = \varepsilon^{\dot{a}\dot{b}} \eta_a^\dagger \zeta_b^\dagger$$

在固有保时向 Lorentz 变换下**不变**，满足

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = [d^*(\Lambda)]_{\dot{a}}^{\dot{b}} \eta_b^\dagger [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_b^\dagger [d^\dagger(\Lambda)]^{\dot{b}}_{\dot{a}} [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_b^\dagger \delta^{\dot{b}}_{\dot{c}} \zeta^{\dagger\dot{c}} = \eta_a^\dagger \zeta^{\dagger\dot{a}}$$

 第二步用了**转置**性质 $[d^*(\Lambda)]_{\dot{a}}^{\dot{b}} = [d^\dagger(\Lambda)]^{\dot{b}}_{\dot{a}}$ ，可见 $\eta_a^\dagger \zeta^{\dagger\dot{a}}$ 是 **Lorentz 标量**

 由 $\eta^{\dagger\dot{1}} = \eta_2^\dagger$ 、 $\eta^{\dagger\dot{2}} = -\eta_1^\dagger$ 、 $\zeta^{\dagger\dot{1}} = \zeta_2^\dagger$ 和 $\zeta^{\dagger\dot{2}} = -\zeta_1^\dagger$ 得

$$\eta_a^\dagger \zeta^{\dagger\dot{a}} = \eta_1^\dagger \zeta^{\dagger\dot{1}} + \eta_2^\dagger \zeta^{\dagger\dot{2}} = -\eta^{\dagger\dot{2}} \zeta^{\dagger\dot{1}} + \eta^{\dagger\dot{1}} \zeta^{\dagger\dot{2}} = -\eta^{\dagger\dot{2}} \zeta_2^\dagger - \eta^{\dagger\dot{1}} \zeta_1^\dagger = -\eta^{\dagger\dot{a}} \zeta_a^\dagger$$

 即参与缩并的**带点旋量指标一升一降**会多出一个**负号**

 假设右手 Weyl 旋量 $\eta^{\dagger\dot{a}}$ 和 $\zeta_a^\dagger$ 都是 **Grassmann 数**，则 $\eta^{\dagger\dot{a}} \zeta_a^\dagger = -\zeta_a^\dagger \eta^{\dagger\dot{a}}$

 将**省略带点旋量指标的内积**写成 $\eta^\dagger \zeta^\dagger \equiv \eta_a^\dagger \zeta^{\dagger\dot{a}} = -\eta^{\dagger\dot{a}} \zeta_a^\dagger = \zeta_a^\dagger \eta^{\dagger\dot{a}} = \zeta^\dagger \eta^\dagger$

 则内积 $\eta^\dagger \zeta^\dagger$ 和 $\zeta^\dagger \eta^\dagger$ **相等**

Lorentz 不变量和 Weyl 旋量算符

🐮 可见，只要将不带点和带点的旋量指标分别缩并完毕，就能得到 Lorentz 标量

👒 另一方面，缩并一个不带点的指标和一个带点的指标并不能得到 Lorentz 不变量

🕶 比如， $\eta^a \zeta_a^\dagger$ 和 $\eta^{\dagger\dot{a}} \zeta_a$ 都不是 Lorentz 标量

Lorentz 不变量和 Weyl 旋量算符

🐮 可见，只要将不带点和带点的旋量指标分别缩并完毕，就能得到 Lorentz 标量

👔 另一方面，缩并一个不带点的指标和一个带点的指标并不能得到 Lorentz 不变量

🕶 比如， $\eta^a \zeta_a^\dagger$ 和 $\eta^{\dagger a} \zeta_a$ 都不是 Lorentz 标量

👔 对于 Weyl 旋量算符 η_a 和 ζ_a ，有

$$(\eta\zeta)^\dagger = (\eta^a \zeta_a)^\dagger = (\zeta_a)^\dagger (\eta^a)^\dagger = \zeta_a^\dagger \eta^{\dagger a} = \zeta^\dagger \eta^\dagger$$

👔 即 $\zeta^\dagger \eta^\dagger$ 是 $\eta\zeta$ 的厄米共轭算符

👔 厄米共轭操作将左手和右手 Weyl 旋量算符相互转换

9.6.2 小节 Dirac 和 Majorana 旋量场的分解

🐘 依照上一小节关于旋量指标的约定，将 Dirac 旋量场 $\psi(x)$ 分解成左手 Weyl 旋量场 $\eta_a(x)$ 和右手 Weyl 旋量场 $\zeta^{\dagger\dot{a}}(x)$ ，形式为

$$\psi(x) = \begin{pmatrix} \eta_a(x) \\ \zeta^{\dagger\dot{a}}(x) \end{pmatrix}$$

🚩 在量子化之前， $\eta_a(x)$ 和 $\zeta^{\dagger\dot{a}}(x)$ 是 Grassmann 数，因而 $\psi(x)$ 也是 Grassmann 数

💡 这是在 9.2.1 小节中转置两个旋量场必须添加一个额外负号的原因

🏰 根据 $D(\Lambda) = \begin{pmatrix} d(\Lambda) & \\ & d^{-1\dagger}(\Lambda) \end{pmatrix}$ ， $\psi(x)$ 的固有保时向 Lorentz 变换表达成

$$\begin{pmatrix} \eta'_a(x') \\ \zeta'^{\dagger\dot{a}}(x') \end{pmatrix} = \psi'(x') = D(\Lambda)\psi(x) = \begin{pmatrix} [d(\Lambda)]^a_b \eta_b(x) \\ [d^{-1\dagger}(\Lambda)]^{\dot{a}}_{\dot{b}} \zeta^{\dagger\dot{b}}(x) \end{pmatrix}$$

🧠 $\psi(x)$ 的 Dirac 共轭是 $\bar{\psi} = \psi^\dagger \gamma^0 = \begin{pmatrix} \eta^\dagger_b & \zeta^b \end{pmatrix} \begin{pmatrix} & \delta^{\dot{b}}_{\dot{a}} \\ \delta_b^a & \end{pmatrix} = \begin{pmatrix} \zeta^a & \eta^\dagger_{\dot{a}} \end{pmatrix}$

Dirac 矩阵的指标形式

 保持旋量指标平衡, 则 Dirac 方程 $(i\gamma^\mu \partial_\mu - m)\psi = 0$ 化为

$$\begin{pmatrix} -m\delta_a{}^b & i(\sigma^\mu)_{a\dot{b}}\partial_\mu \\ i(\bar{\sigma}^\mu)^{\dot{a}b}\partial_\mu & -m\delta^{\dot{a}}{}_{\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b \\ \zeta^{\dagger\dot{b}} \end{pmatrix} = 0$$

 因而 Dirac 矩阵的指标形式是

$$\gamma^\mu = \begin{pmatrix} & (\sigma^\mu)_{a\dot{b}} \\ (\bar{\sigma}^\mu)^{\dot{a}b} & \end{pmatrix}$$

 注意, γ^μ 中的 γ^0 与 Dirac 共轭 $\bar{\psi} = \psi^\dagger \gamma^0 = \begin{pmatrix} \eta_a^\dagger & \zeta^a \end{pmatrix} \begin{pmatrix} & \delta^{\dot{b}}{}_{\dot{a}} \\ \delta_b{}^a & \end{pmatrix} = \begin{pmatrix} \zeta^a & \eta_{\dot{a}}^\dagger \end{pmatrix}$

中的 γ^0 具有不同的指标结构

 两者本质不同, 有些书将后者记为 β 以示区别

σ^μ 和 $\bar{\sigma}^\mu$ 的 Lorentz 变换规则

🐰 于是, γ^μ 的 Lorentz 变换规则 $D^{-1}(\Lambda)\gamma^\mu D(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu$ 左边变成

$$\begin{aligned}
& D^{-1}(\Lambda)\gamma^\mu D(\Lambda) \\
&= \begin{pmatrix} [d^{-1}(\Lambda)]_a{}^c & \\ & [d^\dagger(\Lambda)]^{\dot{a}}{}_{\dot{c}} \end{pmatrix} \begin{pmatrix} (\sigma^\mu)_{cd} \\ (\bar{\sigma}^\mu)^{\dot{c}d} \end{pmatrix} \begin{pmatrix} [d(\Lambda)]_d{}^b & \\ & [d^{-1\dagger}(\Lambda)]^{\dot{d}}{}_{\dot{b}} \end{pmatrix} \\
&= \begin{pmatrix} [d^{-1}(\Lambda)]_a{}^c (\sigma^\mu)_{cd} [d^{-1\dagger}(\Lambda)]^{\dot{d}}{}_{\dot{b}} & \\ [d^\dagger(\Lambda)]^{\dot{a}}{}_{\dot{c}} (\bar{\sigma}^\mu)^{\dot{c}d} [d(\Lambda)]_d{}^b & \end{pmatrix}
\end{aligned}$$

🎵 右边化为

$$\Lambda^\mu{}_\nu \gamma^\nu = \begin{pmatrix} \Lambda^\mu{}_\nu (\sigma^\nu)_{ab} \\ \Lambda^\mu{}_\nu (\bar{\sigma}^\nu)^{\dot{a}\dot{b}} \end{pmatrix}$$

🍊 两相比较, 推出

$$[d^{-1}(\Lambda)]_a{}^c (\sigma^\mu)_{cd} [d^{-1\dagger}(\Lambda)]^{\dot{d}}{}_{\dot{b}} = \Lambda^\mu{}_\nu (\sigma^\nu)_{ab}, \quad [d^\dagger(\Lambda)]^{\dot{a}}{}_{\dot{c}} (\bar{\sigma}^\mu)^{\dot{c}d} [d(\Lambda)]_d{}^b = \Lambda^\mu{}_\nu (\bar{\sigma}^\nu)^{\dot{a}\dot{b}}$$

🌲 这分别是 σ^μ 和 $\bar{\sigma}^\mu$ 的 Lorentz 变换规则, 可将它们看作两个 Lorentz 矢量

Lorentz 矢量 $\eta\sigma^\mu\zeta^\dagger$ 和 $\eta^\dagger\bar{\sigma}^\mu\zeta$



对任意 Weyl 旋量 η 和 ζ ，定义

$$\eta\sigma^\mu\zeta^\dagger \equiv \eta^a(\sigma^\mu)_{ab}\zeta^{\dagger b}, \quad \eta^\dagger\bar{\sigma}^\mu\zeta \equiv \eta^\dagger_{\dot{a}}(\bar{\sigma}^\mu)^{\dot{a}b}\zeta_b$$



它们都是 Lorentz 矢量，相应的固有保时向 Lorentz 变换为

$$\begin{aligned} \eta'\sigma^\mu\zeta'^\dagger &= [d^{-1T}(\Lambda)]^a{}_c\eta^c(\sigma^\mu)_{ab}[d^{-1\dagger}(\Lambda)]^{\dot{b}}{}_{\dot{d}}\zeta^{\dagger\dot{d}} = \eta^c[d^{-1}(\Lambda)]_c{}^a(\sigma^\mu)_{ab}[d^{-1\dagger}(\Lambda)]^{\dot{b}}{}_{\dot{d}}\zeta^{\dagger\dot{d}} \\ &= \eta^c\Lambda^\mu{}_\nu(\sigma^\nu)_{cd}\zeta^{\dagger d} = \Lambda^\mu{}_\nu\eta\sigma^\nu\zeta^\dagger \\ \eta'^\dagger\bar{\sigma}^\mu\zeta' &= [d^*(\Lambda)]_{\dot{a}}{}^{\dot{c}}\eta^\dagger_{\dot{c}}(\bar{\sigma}^\mu)^{\dot{a}b}[d(\Lambda)]_b{}^d\zeta_d = \eta^\dagger_{\dot{c}}[d^\dagger(\Lambda)]^{\dot{c}}{}_{\dot{a}}(\bar{\sigma}^\mu)^{\dot{a}b}[d(\Lambda)]_b{}^d\zeta_d \\ &= \eta^\dagger_{\dot{c}}\Lambda^\mu{}_\nu(\bar{\sigma}^\nu)^{\dot{c}d}\zeta_d = \Lambda^\mu{}_\nu\eta^\dagger\bar{\sigma}^\nu\zeta \end{aligned}$$

Lorentz 矢量 $\eta\sigma^\mu\zeta^\dagger$ 和 $\eta^\dagger\bar{\sigma}^\mu\zeta$ 对任意 Weyl 旋量 η 和 ζ ，定义

$$\eta\sigma^\mu\zeta^\dagger \equiv \eta^a(\sigma^\mu)_{ab}\zeta^{\dagger b}, \quad \eta^\dagger\bar{\sigma}^\mu\zeta \equiv \eta^\dagger_a(\bar{\sigma}^\mu)^{\dot{a}b}\zeta_b$$



它们都是 Lorentz 矢量，相应的固有保时向 Lorentz 变换为

$$\begin{aligned}\eta'\sigma^\mu\zeta'^\dagger &= [d^{-1T}(\Lambda)]^a{}_c\eta^c(\sigma^\mu)_{ab}[d^{-1\dagger}(\Lambda)]^{\dot{b}}{}_{\dot{d}}\zeta^{\dagger\dot{d}} = \eta^c[d^{-1}(\Lambda)]_c{}^a(\sigma^\mu)_{ab}[d^{-1\dagger}(\Lambda)]^{\dot{b}}{}_{\dot{d}}\zeta^{\dagger\dot{d}} \\ &= \eta^c\Lambda^\mu{}_\nu(\sigma^\nu)_{cd}\zeta^{\dagger\dot{d}} = \Lambda^\mu{}_\nu\eta\sigma^\nu\zeta^\dagger\end{aligned}$$

$$\begin{aligned}\eta'^\dagger\bar{\sigma}^\mu\zeta' &= [d^*(\Lambda)]_{\dot{a}}{}^{\dot{c}}\eta^\dagger_{\dot{c}}(\bar{\sigma}^\mu)^{\dot{a}b}[d(\Lambda)]_b{}^d\zeta_d = \eta^\dagger_{\dot{c}}[d^\dagger(\Lambda)]^{\dot{c}}{}_{\dot{a}}(\bar{\sigma}^\mu)^{\dot{a}b}[d(\Lambda)]_b{}^d\zeta_d \\ &= \eta^\dagger_{\dot{c}}\Lambda^\mu{}_\nu(\bar{\sigma}^\nu)^{\dot{c}d}\zeta_d = \Lambda^\mu{}_\nu\eta^\dagger\bar{\sigma}^\nu\zeta\end{aligned}$$

由 $\sigma^2\sigma^\mu\sigma^2 = (\bar{\sigma}^\mu)^T$ 得 $(i\sigma^2)\sigma^\mu(i\sigma^2) = -(\bar{\sigma}^\mu)^T$ ，相应的指标形式为

$$\varepsilon^{ac}(\sigma^\mu)_{cd}\varepsilon^{\dot{d}b} = -[(\bar{\sigma}^\mu)^T]^{ab} = -(\bar{\sigma}^\mu)^{ba}$$

对于 Weyl 旋量场 $\eta_a(x)$ 和 $\zeta^{\dagger\dot{a}}(x)$ ，有

Grassmann 数性质

$$\begin{aligned}[\eta^a(\sigma^\mu)_{ab}\zeta^{\dagger\dot{b}}]^\dagger &= \zeta^b(\sigma^\mu)_{ba}\eta^{\dagger\dot{a}} = -\eta^{\dagger\dot{a}}(\sigma^\mu)_{ba}\zeta^b = -\varepsilon^{\dot{a}c}\eta^\dagger_{\dot{c}}(\sigma^\mu)_{ba}\varepsilon^{bd}\zeta_d \\ &= \eta^\dagger_{\dot{c}}\varepsilon^{db}(\sigma^\mu)_{ba}\varepsilon^{\dot{a}c}\zeta_d = -\eta^\dagger_{\dot{c}}(\bar{\sigma}^\mu)^{\dot{c}d}\zeta_d = -[\zeta^\dagger_{\dot{d}}(\bar{\sigma}^\mu)^{\dot{d}c}\eta_c]^\dagger\end{aligned}$$



即

$$(\eta\sigma^\mu\zeta^\dagger)^\dagger = \zeta\sigma^\mu\eta^\dagger = -\eta^\dagger\bar{\sigma}^\mu\zeta = -(\zeta^\dagger\bar{\sigma}^\mu\eta)^\dagger$$

Lorentz 张量 $\eta\sigma^\mu\bar{\sigma}^\nu\zeta$ 和 $\eta^\dagger\bar{\sigma}^\mu\sigma^\nu\zeta^\dagger$

🐔 类似地, $\eta\sigma^\mu\bar{\sigma}^\nu\zeta \equiv \eta^a(\sigma^\mu)_{ab}(\bar{\sigma}^\nu)^{bc}\zeta_c$ 和 $\eta^\dagger\bar{\sigma}^\mu\sigma^\nu\zeta^\dagger \equiv \eta_a^\dagger(\bar{\sigma}^\mu)^{\dot{a}b}(\sigma^\nu)_{b\dot{c}}\zeta^{\dagger\dot{c}}$ 都是二阶 Lorentz 张量

📦 由 $\sigma^2\bar{\sigma}^\mu\sigma^2 = (\sigma^\mu)^T$ 得 $(-i\sigma^2)\bar{\sigma}^\mu(-i\sigma^2) = -(\sigma^\mu)^T$, 相应的指标形式为

$$\varepsilon_{\dot{a}\dot{c}}(\bar{\sigma}^\mu)^{\dot{c}d}\varepsilon_{db} = -[(\sigma^\mu)^T]_{\dot{a}b} = -(\sigma^\mu)_{b\dot{a}}$$

💎 再利用 $\varepsilon_{ab}\varepsilon^{bc} = \delta_a^c$ 和 $\varepsilon^{ac}(\sigma^\mu)_{cd}\varepsilon^{\dot{d}b} = -[(\bar{\sigma}^\mu)^T]^{ab} = -(\bar{\sigma}^\mu)^{ba}$ 推出

$$\begin{aligned}\varepsilon_{\dot{a}\dot{c}}(\bar{\sigma}^\nu)^{\dot{c}d}(\sigma^\mu)_{d\dot{e}}\varepsilon^{\dot{e}b} &= \varepsilon_{\dot{a}\dot{c}}(\bar{\sigma}^\nu)^{\dot{c}d}\delta_{\dot{a}}^f(\sigma^\mu)_{f\dot{e}}\varepsilon^{\dot{e}b} = \varepsilon_{\dot{a}\dot{c}}(\bar{\sigma}^\nu)^{\dot{c}d}\varepsilon_{d\dot{g}}\varepsilon^{g\dot{f}}(\sigma^\mu)_{f\dot{e}}\varepsilon^{\dot{e}b} \\ &= (-\sigma^\nu)_{g\dot{a}}(-\bar{\sigma}^\mu)^{\dot{b}g} = (\bar{\sigma}^\mu)^{\dot{b}g}(\sigma^\nu)_{g\dot{a}}\end{aligned}$$

🔲 故 $[\eta^a(\sigma^\mu)_{ab}(\bar{\sigma}^\nu)^{bc}\zeta_c]^\dagger = \zeta_c^\dagger(\bar{\sigma}^\nu)^{\dot{c}b}(\sigma^\mu)_{b\dot{a}}\eta^{\dagger\dot{a}} = -\eta^{\dagger\dot{a}}(\bar{\sigma}^\nu)^{\dot{c}b}(\sigma^\mu)_{b\dot{a}}\zeta_c^\dagger$
 $= -\varepsilon^{\dot{a}d}\eta_d^\dagger(\bar{\sigma}^\nu)^{\dot{c}b}(\sigma^\mu)_{b\dot{a}}\varepsilon_{\dot{c}\dot{e}}\zeta^{\dagger\dot{e}} = \eta_d^\dagger\varepsilon_{\dot{c}\dot{e}}(\bar{\sigma}^\nu)^{\dot{c}b}(\sigma^\mu)_{b\dot{a}}\varepsilon^{\dot{a}d}\zeta^{\dagger\dot{e}}$
 $= \eta_d^\dagger(\bar{\sigma}^\mu)^{\dot{d}g}(\sigma^\nu)_{g\dot{e}}\zeta^{\dagger\dot{e}} = [\zeta^e(\sigma^\nu)_{e\dot{g}}(\bar{\sigma}^\mu)^{\dot{g}d}\eta_d]^\dagger$

💍 即 $(\eta\sigma^\mu\bar{\sigma}^\nu\zeta)^\dagger = \zeta^\dagger\bar{\sigma}^\nu\sigma^\mu\eta^\dagger = \eta^\dagger\bar{\sigma}^\mu\sigma^\nu\zeta^\dagger = (\zeta\sigma^\nu\bar{\sigma}^\mu\eta)^\dagger$

旋量双线性型的分解



将 **Dirac 旋量双线性型** 分解成由 **Weyl 旋量** 表达的 **Lorentz 张量**，有

$$\bar{\psi}\psi = \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} \eta_a \\ \zeta^{\dagger\dot{a}} \end{pmatrix} = \zeta^a \eta_a + \eta_a^\dagger \zeta^{\dagger\dot{a}} = \zeta \eta + \eta^\dagger \zeta^\dagger$$

$$\bar{\psi}\gamma^5\psi = \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} -\delta_a^b & \\ & \delta^{\dot{a}}_{\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b \\ \zeta^{\dagger\dot{b}} \end{pmatrix} = -\zeta^a \eta_a + \eta_a^\dagger \zeta^{\dagger\dot{a}} = -\zeta \eta + \eta^\dagger \zeta^\dagger$$

$$\begin{aligned} \bar{\psi}\gamma^\mu\psi &= \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} & (\sigma^\mu)_{a\dot{b}} \\ (\bar{\sigma}^\mu)^{\dot{a}b} & \end{pmatrix} \begin{pmatrix} \eta_b \\ \zeta^{\dagger\dot{b}} \end{pmatrix} = \zeta^a (\sigma^\mu)_{a\dot{b}} \zeta^{\dagger\dot{b}} + \eta_a^\dagger (\bar{\sigma}^\mu)^{\dot{a}b} \eta_b \\ &= \zeta \sigma^\mu \zeta^\dagger + \eta^\dagger \bar{\sigma}^\mu \eta \end{aligned}$$

$$\begin{aligned} \bar{\psi}\gamma^\mu\gamma^5\psi &= \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} & (\sigma^\mu)_{a\dot{b}} \\ (\bar{\sigma}^\mu)^{\dot{a}b} & \end{pmatrix} \begin{pmatrix} -\delta_b^c & \\ & \delta^{\dot{b}}_{\dot{c}} \end{pmatrix} \begin{pmatrix} \eta_c \\ \zeta^{\dagger\dot{c}} \end{pmatrix} \\ &= \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} & (\sigma^\mu)_{a\dot{b}} \\ (\bar{\sigma}^\mu)^{\dot{a}b} & \end{pmatrix} \begin{pmatrix} -\eta_b \\ \zeta^{\dagger\dot{b}} \end{pmatrix} = \zeta^a (\sigma^\mu)_{a\dot{b}} \zeta^{\dagger\dot{b}} - \eta_a^\dagger (\bar{\sigma}^\mu)^{\dot{a}b} \eta_b \\ &= \zeta \sigma^\mu \zeta^\dagger - \eta^\dagger \bar{\sigma}^\mu \eta \end{aligned}$$

旋量双线性型的分解

 还有

$$\begin{aligned}
\bar{\psi}\sigma^{\mu\nu}\psi &= \frac{i}{2} \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} (\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu)_a{}^b & \\ & (\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu)^{\dot{a}}{}_{\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b \\ \zeta^{\dagger\dot{b}} \end{pmatrix} \\
&= \frac{i}{2} \zeta^a (\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu)_a{}^b \eta_b + \frac{i}{2} \eta_a^\dagger (\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu)^{\dot{a}}{}_{\dot{b}} \zeta^{\dagger\dot{b}} \\
&= \frac{i}{2} \zeta (\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu) \eta + \frac{i}{2} \eta^\dagger (\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu) \zeta^\dagger
\end{aligned}$$

 进一步推出

$$\begin{aligned}
\overline{\psi_R}\psi_L &= \frac{1}{2} \bar{\psi}(1 - \gamma^5)\psi = \zeta\eta \\
\overline{\psi_L}\psi_R &= \frac{1}{2} \bar{\psi}(1 + \gamma^5)\psi = \eta^\dagger\zeta^\dagger \\
\overline{\psi_L}\gamma^\mu\psi_L &= \frac{1}{2} \bar{\psi}(\gamma^\mu - \gamma^\mu\gamma^5)\psi = \eta^\dagger\bar{\sigma}^\mu\eta \\
\overline{\psi_R}\gamma^\mu\psi_R &= \frac{1}{2} \bar{\psi}(\gamma^\mu + \gamma^\mu\gamma^5)\psi = \zeta\sigma^\mu\zeta^\dagger
\end{aligned}$$

拉氏量的分解

🕊 另一方面，自由 Dirac 旋量场的拉氏量分解为

$$\begin{aligned}\mathcal{L} &= \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = \begin{pmatrix} \zeta^a & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} -m\delta_a^b & i(\sigma^\mu)_{ab}\partial_\mu \\ i(\bar{\sigma}^\mu)^{\dot{a}b}\partial_\mu & -m\delta^{\dot{a}}_{\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b \\ \zeta^{\dagger\dot{b}} \end{pmatrix} \\ &= -m\zeta^a\eta_a + i\zeta^a(\sigma^\mu)_{ab}\partial_\mu\zeta^{\dagger\dot{b}} + i\eta_a^\dagger(\bar{\sigma}^\mu)^{\dot{a}b}\partial_\mu\eta_b - m\eta_a^\dagger\zeta^{\dagger\dot{a}} \\ &= i\eta^\dagger\bar{\sigma}^\mu\partial_\mu\eta + i\zeta\sigma^\mu\partial_\mu\zeta^\dagger - m(\zeta\eta + \eta^\dagger\zeta^\dagger)\end{aligned}$$

🌐 这里的质量项涉及两个不同的 Weyl 旋量场 $\eta_a(x)$ 和 $\zeta_a(x)$ ，称为 Dirac 质量项

🎨 如果质量 $m = 0$ ，则

$$\mathcal{L}_L = i\eta^\dagger\bar{\sigma}^\mu\partial_\mu\eta \quad \text{和} \quad \mathcal{L}_R = i\zeta\sigma^\mu\partial_\mu\zeta^\dagger$$

分别描述自由的左手 Weyl 旋量场 $\eta_a(x)$ 和右手 Weyl 旋量场 $\zeta^{\dagger\dot{a}}(x)$

🏡 相应的运动方程是两个 Weyl 方程：

$$i\bar{\sigma}^\mu\partial_\mu\eta = 0, \quad i\sigma^\mu\partial_\mu\zeta^\dagger = 0$$

Weyl 旋量场的 C 变换

🐘 下面讨论 **Weyl 旋量场的分立变换**

🧸 首先，**电荷共轭矩阵**的指标形式为 $C = \begin{pmatrix} -i\sigma^2 & \\ & i\sigma^2 \end{pmatrix} = \begin{pmatrix} \varepsilon_{ab} & \\ & \varepsilon^{\dot{a}\dot{b}} \end{pmatrix}$

🧩 将 $\psi(x)$ 的**电荷共轭场** $\psi^C(x)$ 分解成 Weyl 旋量场，得到

$$\psi^C(x) = C\bar{\psi}^T(x) = C \begin{pmatrix} \zeta^b(x) & \eta_b^\dagger(x) \end{pmatrix}^T = \begin{pmatrix} \varepsilon_{ab} & \\ & \varepsilon^{\dot{a}\dot{b}} \end{pmatrix} \begin{pmatrix} \zeta^b(x) \\ \eta_b^\dagger(x) \end{pmatrix} = \begin{pmatrix} \zeta_a(x) \\ \eta^{\dagger\dot{a}}(x) \end{pmatrix}$$

♟ 从而，**Dirac 旋量场** $\psi(x)$ 的 **C 变换**化为

$$\begin{pmatrix} C^{-1}\eta_a(x)C \\ C^{-1}\zeta^{\dagger\dot{a}}(x)C \end{pmatrix} = C^{-1}\psi(x)C = \zeta_C^*\psi^C(x) = \begin{pmatrix} \zeta_C^*\zeta_a(x) \\ \zeta_C^*\eta^{\dagger\dot{a}}(x) \end{pmatrix}$$

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$$\psi^C(x) = C\bar{\psi}^T(x) = C \begin{pmatrix} \zeta^b(x) & \eta_b^\dagger(x) \end{pmatrix}^T = \begin{pmatrix} \varepsilon_{ab} & \\ & \varepsilon^{\dot{a}\dot{b}} \end{pmatrix} \begin{pmatrix} \zeta^b(x) \\ \eta_b^\dagger(x) \end{pmatrix} = \begin{pmatrix} \zeta_a(x) \\ \eta^{\dagger\dot{a}}(x) \end{pmatrix}$$

👤 从而，**Dirac 旋量场** $\psi(x)$ 的 C 变换化为

$$\begin{pmatrix} C^{-1}\eta_a(x)C \\ C^{-1}\zeta^{\dagger\dot{a}}(x)C \end{pmatrix} = C^{-1}\psi(x)C = \zeta_C^*\psi^C(x) = \begin{pmatrix} \zeta_C^*\zeta_a(x) \\ \zeta_C^*\eta^{\dagger\dot{a}}(x) \end{pmatrix}$$

📌 即**左右手** Weyl 旋量场的 C 变换是

$$C^{-1}\eta_a(x)C = \zeta_C^*\zeta_a(x), \quad C^{-1}\zeta^{\dagger\dot{a}}(x)C = \zeta_C^*\eta^{\dagger\dot{a}}(x)$$

🎲 可见，电荷共轭变换将 η 和 ζ 相互转换。取厄米共轭，得 $C^{-1}\eta_b^\dagger(x)C = \zeta_C\zeta_b^\dagger(x)$ 及 $C^{-1}\zeta^b(x)C = \zeta_C\eta^b(x)$ ，分别与 $\varepsilon^{\dot{a}\dot{b}}$ 和 ε_{ab} 缩并，推出

$$C^{-1}\eta^{\dagger\dot{a}}(x)C = \zeta_C\zeta^{\dagger\dot{a}}(x), \quad C^{-1}\zeta_a(x)C = \zeta_C\eta_a(x)$$

Weyl 旋量场的 P 变换

🐪 其次，Dirac 旋量场 $\psi(x)$ 的 P 变换表达为

$$\begin{aligned} \begin{pmatrix} P^{-1}\eta_a(x)P \\ P^{-1}\zeta^{\dagger\dot{a}}(x)P \end{pmatrix} &= P^{-1}\psi(x)P = \zeta_P^* \gamma^0 \psi(\mathcal{P}x) \\ &= \zeta_P^* \begin{pmatrix} \delta^{\dot{a}}_{\dot{b}} \\ \delta_a^b \end{pmatrix} \begin{pmatrix} \eta_b(\mathcal{P}x) \\ \zeta^{\dagger\dot{b}}(\mathcal{P}x) \end{pmatrix} = \begin{pmatrix} \zeta_P^* \zeta^{\dagger\dot{a}}(\mathcal{P}x) \\ \zeta_P^* \eta_a(\mathcal{P}x) \end{pmatrix} \end{aligned}$$

🐼 注意此处 γ^0 的指标结构与 $\bar{\psi} = \psi^\dagger \gamma^0$ 中一样

Weyl 旋量场的 P 变换

🐪 其次, Dirac 旋量场 $\psi(x)$ 的 P 变换表达为

$$\begin{aligned} \begin{pmatrix} P^{-1}\eta_a(x)P \\ P^{-1}\zeta^{\dagger\dot{a}}(x)P \end{pmatrix} &= P^{-1}\psi(x)P = \zeta_P^*\gamma^0\psi(\mathcal{P}x) \\ &= \zeta_P^* \begin{pmatrix} & \delta^{\dot{a}}_{\dot{b}} \\ \delta_a^b & \end{pmatrix} \begin{pmatrix} \eta_b(\mathcal{P}x) \\ \zeta^{\dagger\dot{b}}(\mathcal{P}x) \end{pmatrix} = \begin{pmatrix} \zeta_P^*\zeta^{\dagger\dot{a}}(\mathcal{P}x) \\ \zeta_P^*\eta_a(\mathcal{P}x) \end{pmatrix} \end{aligned}$$

♠ 注意此处 γ^0 的指标结构与 $\bar{\psi} = \psi^\dagger\gamma^0$ 中一样

♠ 于是得到左右手 Weyl 旋量场的 P 变换

$$P^{-1}\eta_a(x)P = \zeta_P^*\zeta^{\dagger\dot{a}}(\mathcal{P}x), \quad P^{-1}\zeta^{\dagger\dot{a}}(x)P = \zeta_P^*\eta_a(\mathcal{P}x)$$

♥ 也就是说, 宇称变换将左手和右手 Weyl 旋量场相互转换

♣ 取厄米共轭得 $P^{-1}\eta_b^\dagger(x)P = \zeta_P\zeta^b(\mathcal{P}x)$ 和 $P^{-1}\zeta^b(x)P = \zeta_P\eta_b^\dagger(\mathcal{P}x)$

♦ 两边与 $i\sigma^2 = \varepsilon^{\dot{a}\dot{b}} = -\varepsilon_{ab}$ 缩并, 推出

$$P^{-1}\eta^{\dagger\dot{a}}(x)P = -\zeta_P\zeta_a(\mathcal{P}x), \quad P^{-1}\zeta_a(x)P = -\zeta_P\eta^{\dagger\dot{a}}(\mathcal{P}x)$$

Weyl 旋量场的 T 变换

🐪 最后，时间反演矩阵的指标形式是 $D(\mathcal{T}) = \mathcal{C}\gamma^5 = \begin{pmatrix} i\sigma^2 & \\ & i\sigma^2 \end{pmatrix} = \begin{pmatrix} \varepsilon^{ab} & \\ & -\varepsilon_{\dot{a}\dot{b}} \end{pmatrix}$

🎯 Dirac 旋量场 $\psi(x)$ 的 T 变换化为

$$\begin{aligned} \begin{pmatrix} T^{-1}\eta_a(x)T \\ T^{-1}\zeta^{\dagger\dot{a}}(x)T \end{pmatrix} &= T^{-1}\psi(x)T = \zeta_T^* \mathcal{C}\gamma^5 \psi(\mathcal{T}x) \\ &= \zeta_T^* \begin{pmatrix} \varepsilon^{ab} & \\ & -\varepsilon_{\dot{a}\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b(\mathcal{T}x) \\ \zeta^{\dagger\dot{b}}(\mathcal{T}x) \end{pmatrix} = \begin{pmatrix} \zeta_T^* \eta^a(\mathcal{T}x) \\ -\zeta_T^* \zeta_{\dot{a}}^{\dagger}(\mathcal{T}x) \end{pmatrix} \end{aligned}$$

Weyl 旋量场的 T 变换

🐪 最后，时间反演矩阵的指标形式是 $D(\mathcal{T}) = \mathcal{C}\gamma^5 = \begin{pmatrix} i\sigma^2 & \\ & i\sigma^2 \end{pmatrix} = \begin{pmatrix} \varepsilon^{ab} & \\ & -\varepsilon_{\dot{a}\dot{b}} \end{pmatrix}$

🎯 Dirac 旋量场 $\psi(x)$ 的 T 变换化为

$$\begin{aligned} \begin{pmatrix} T^{-1}\eta_a(x)T \\ T^{-1}\zeta^{\dagger\dot{a}}(x)T \end{pmatrix} &= T^{-1}\psi(x)T = \zeta_T^* \mathcal{C}\gamma^5 \psi(\mathcal{T}x) \\ &= \zeta_T^* \begin{pmatrix} \varepsilon^{ab} & \\ & -\varepsilon_{\dot{a}\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b(\mathcal{T}x) \\ \zeta^{\dagger\dot{b}}(\mathcal{T}x) \end{pmatrix} = \begin{pmatrix} \zeta_T^* \eta^a(\mathcal{T}x) \\ -\zeta_T^* \zeta_{\dot{a}}^{\dagger}(\mathcal{T}x) \end{pmatrix} \end{aligned}$$

🔮 则左右手 Weyl 旋量场的 T 变换是

$$T^{-1}\eta_a(x)T = \zeta_T^* \eta^a(\mathcal{T}x), \quad T^{-1}\zeta^{\dagger\dot{a}}(x)T = -\zeta_T^* \zeta_{\dot{a}}^{\dagger}(\mathcal{T}x)$$

🌟 取厄米共轭，有 $T^{-1}\eta_b^{\dagger}(x)T = \zeta_T \eta^{\dagger\dot{b}}(\mathcal{T}x)$ 和 $T^{-1}\zeta^b(x)T = -\zeta_T \zeta_b(\mathcal{T}x)$

🧙 与 $i\sigma^2 = \varepsilon^{\dot{a}\dot{b}} = -\varepsilon_{\dot{a}\dot{b}} = -\varepsilon_{ab} = \varepsilon^{ab}$ 缩并，得

$$T^{-1}\eta^{\dagger\dot{a}}(x)T = -\zeta_T \eta_{\dot{a}}^{\dagger}(\mathcal{T}x), \quad T^{-1}\zeta_a(x)T = \zeta_T \zeta^a(\mathcal{T}x)$$

Majorana 旋量场的分解



下面讨论 **Majorana 旋量场**，**Majorana 条件**意味着
$$\begin{pmatrix} \eta_a \\ \zeta^{\dagger\dot{a}} \end{pmatrix} = \psi = \mathcal{C}\bar{\psi}^T = \begin{pmatrix} \zeta_a \\ \eta^{\dagger\dot{a}} \end{pmatrix}$$



即 $\eta = \zeta$ ，这表明 Majorana 旋量场中的**左手**和**右手** Weyl 旋量场是**相关的**



因此，可以将 Majorana 旋量场 $\psi(x)$ 分解成

$$\psi(x) = \begin{pmatrix} \eta_a(x) \\ \eta^{\dagger\dot{a}}(x) \end{pmatrix}$$

Majorana 旋量场的分解

 下面讨论 **Majorana 旋量场**，**Majorana 条件**意味着 $\begin{pmatrix} \eta_a \\ \zeta^{\dagger\dot{a}} \end{pmatrix} = \psi = \mathcal{C}\bar{\psi}^T = \begin{pmatrix} \zeta_a \\ \eta^{\dagger\dot{a}} \end{pmatrix}$

 即 $\eta = \zeta$ ，这表明 Majorana 旋量场中的**左手**和**右手** Weyl 旋量场是**相关的**

 因此，可以将 Majorana 旋量场 $\psi(x)$ 分解成

$$\psi(x) = \begin{pmatrix} \eta_a(x) \\ \eta^{\dagger\dot{a}}(x) \end{pmatrix}$$

 而**自由 Majorana 旋量场**的**拉氏量**分解为

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = \frac{1}{2} \begin{pmatrix} \eta^a & \eta_{\dot{a}}^\dagger \end{pmatrix} \begin{pmatrix} -m\delta_a^b & i(\sigma^\mu)_{a\dot{b}}\partial_\mu \\ i(\bar{\sigma}^\mu)^{\dot{a}b}\partial_\mu & -m\delta^{\dot{a}}_{\dot{b}} \end{pmatrix} \begin{pmatrix} \eta_b \\ \eta^{\dagger\dot{b}} \end{pmatrix} \\ &= \frac{1}{2} [i\eta^\dagger \bar{\sigma}^\mu \partial_\mu \eta + i\eta \sigma^\mu \partial_\mu \eta^\dagger - m(\eta\eta + \eta^\dagger \eta^\dagger)] \end{aligned}$$

⑧ 利用 $\zeta \sigma^\mu \eta^\dagger = -\eta^\dagger \bar{\sigma}^\mu \zeta$ 将方括号中**第二项**化为

$$i\eta \sigma^\mu \partial_\mu \eta^\dagger = i\partial_\mu (\eta \sigma^\mu \eta^\dagger) - i(\partial_\mu \eta) \sigma^\mu \eta^\dagger = i\partial_\mu (\eta \sigma^\mu \eta^\dagger) + i\eta^\dagger \bar{\sigma}^\mu \partial_\mu \eta$$

 扔掉全散度项 $i\partial_\mu (\eta \sigma^\mu \eta^\dagger)$ ，拉氏量变成 $\mathcal{L} = i\eta^\dagger \bar{\sigma}^\mu \partial_\mu \eta - \frac{1}{2} m(\eta\eta + \eta^\dagger \eta^\dagger)$

 这里的质量项只涉及**一个 Weyl 旋量场** $\eta_a(x)$ ，称为 **Majorana 质量项**

Majorana 旋量场的 $\bar{\psi}\gamma^\mu\psi$ 和 $\bar{\psi}\sigma^{\mu\nu}\psi$

🦊 $\zeta\sigma^\mu\eta^\dagger = -\eta^\dagger\bar{\sigma}^\mu\zeta$ 、 $\eta\sigma^\mu\bar{\sigma}^\nu\zeta = \zeta\sigma^\nu\bar{\sigma}^\mu\eta$ 和 $\eta^\dagger\bar{\sigma}^\mu\sigma^\nu\zeta^\dagger = \zeta^\dagger\bar{\sigma}^\nu\sigma^\mu\eta^\dagger$ 意味着

$$\eta\sigma^\mu\eta^\dagger = -\eta^\dagger\bar{\sigma}^\mu\eta, \quad \eta\sigma^\mu\bar{\sigma}^\nu\eta = \eta\sigma^\nu\bar{\sigma}^\mu\eta, \quad \eta^\dagger\bar{\sigma}^\mu\sigma^\nu\eta^\dagger = \eta^\dagger\bar{\sigma}^\nu\sigma^\mu\eta^\dagger$$

🧡 对于 Majorana 旋量场, $\eta = \zeta$, $\bar{\psi}\gamma^\mu\psi = \zeta\sigma^\mu\zeta^\dagger + \eta^\dagger\bar{\sigma}^\mu\eta$ 化为

$$\bar{\psi}\gamma^\mu\psi = \eta\sigma^\mu\eta^\dagger + \eta^\dagger\bar{\sigma}^\mu\eta = -\eta^\dagger\bar{\sigma}^\mu\eta + \eta^\dagger\bar{\sigma}^\mu\eta = 0$$

🌀 $\bar{\psi}\sigma^{\mu\nu}\psi = \frac{i}{2}\zeta(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu)\eta + \frac{i}{2}\eta^\dagger(\bar{\sigma}^\mu\sigma^\nu - \bar{\sigma}^\nu\sigma^\mu)\zeta^\dagger$ 化为

$$\bar{\psi}\sigma^{\mu\nu}\psi = \frac{i}{2}(\eta\sigma^\mu\bar{\sigma}^\nu\eta - \eta\sigma^\nu\bar{\sigma}^\mu\eta) + \frac{i}{2}(\eta^\dagger\bar{\sigma}^\mu\sigma^\nu\eta^\dagger - \eta^\dagger\bar{\sigma}^\nu\sigma^\mu\eta^\dagger) = 0$$

🧊 这样就验证了 9.2.2 小节的结论

9.6.3 小节 手征旋量场



本小节从手征旋量场的角度分析 Dirac 质量项和 Majorana 质量项的构造过程



用 Weyl 旋量场 $\eta_a(x)$ 和 $\zeta^{\dagger\dot{a}}(x)$ 将四分量左手旋量场 $\psi_L(x)$ 和右手旋量场 $\psi_R(x)$ 表达为

$$\psi_L = \begin{pmatrix} \eta_a \\ 0 \end{pmatrix}, \quad \psi_R = \begin{pmatrix} 0 \\ \zeta^{\dagger\dot{a}} \end{pmatrix}$$

$$\overline{\psi_L} = (\psi_L)^\dagger \gamma^0 = \begin{pmatrix} \eta_b^\dagger & 0 \end{pmatrix} \begin{pmatrix} \delta^{\dot{b}}_{\dot{a}} \\ \delta_b^a \end{pmatrix} = \begin{pmatrix} 0 & \eta_a^\dagger \end{pmatrix}$$

$$\overline{\psi_R} = (\psi_R)^\dagger \gamma^0 = \begin{pmatrix} 0 & \zeta^b \end{pmatrix} \begin{pmatrix} \delta^{\dot{b}}_{\dot{a}} \\ \delta_b^a \end{pmatrix} = \begin{pmatrix} \zeta^a & 0 \end{pmatrix}$$



从而，拉氏量

$$\mathcal{L}_L = i\eta^\dagger \bar{\sigma}^\mu \partial_\mu \eta = i \begin{pmatrix} 0 & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} (\sigma^\mu)_{a\dot{b}} \\ (\bar{\sigma}^\mu)^{\dot{a}b} \end{pmatrix} \partial_\mu \begin{pmatrix} \eta_b \\ 0 \end{pmatrix} = i\overline{\psi_L} \gamma^\mu \partial_\mu \psi_L$$

$$\mathcal{L}_R = i\zeta \sigma^\mu \partial_\mu \zeta^\dagger = i \begin{pmatrix} \zeta^a & 0 \end{pmatrix} \begin{pmatrix} (\sigma^\mu)_{a\dot{b}} \\ (\bar{\sigma}^\mu)^{\dot{a}b} \end{pmatrix} \partial_\mu \begin{pmatrix} 0 \\ \zeta^{\dagger\dot{b}} \end{pmatrix} = i\overline{\psi_R} \gamma^\mu \partial_\mu \psi_R$$



分别描述自由的无质量左手旋量场 ψ_L 和右手旋量场 ψ_R

构造 Dirac 质量项

🐼 如果要构造 **Dirac 质量项**，需采用前面给出的 $\overline{\psi_R}\psi_L = \zeta\eta$ 和 $\overline{\psi_L}\psi_R = \eta^\dagger\zeta^\dagger$ ，得到**自由 Dirac 旋量场** $\psi(x) = \psi_L(x) + \psi_R(x)$ 的拉氏量

$$\begin{aligned}\mathcal{L}_D &= i\eta^\dagger\bar{\sigma}^\mu\partial_\mu\eta + i\zeta\sigma^\mu\partial_\mu\zeta^\dagger - m(\zeta\eta + \eta^\dagger\zeta^\dagger) \\ &= i\overline{\psi_L}\gamma^\mu\partial_\mu\psi_L + i\overline{\psi_R}\gamma^\mu\partial_\mu\psi_R - m(\overline{\psi_R}\psi_L + \overline{\psi_L}\psi_R)\end{aligned}$$

🍏 如果要构造 **Majorana 质量项**，需用到 ψ_L 和 ψ_R 的**电荷共轭场**：

$$(\psi_L)^C = C(\overline{\psi_L})^T = C\begin{pmatrix} 0 & \eta_b^\dagger \end{pmatrix}^T = \begin{pmatrix} \varepsilon_{ab} & \\ & \varepsilon^{\dot{a}b} \end{pmatrix} \begin{pmatrix} 0 \\ \eta_b^\dagger \end{pmatrix} = \begin{pmatrix} 0 \\ \eta^{\dagger\dot{a}} \end{pmatrix} = P_R\psi^C = (\psi^C)_R$$

$$(\psi_R)^C = C(\overline{\psi_R})^T = C\begin{pmatrix} \zeta^b & 0 \end{pmatrix}^T = \begin{pmatrix} \varepsilon_{ab} & \\ & \varepsilon^{\dot{a}b} \end{pmatrix} \begin{pmatrix} \zeta^b \\ 0 \end{pmatrix} = \begin{pmatrix} \zeta_a \\ 0 \end{pmatrix} = P_L\psi^C = (\psi^C)_L$$

$$\overline{(\psi_L)^C} = [(\psi_L)^C]^\dagger\gamma^0 = \begin{pmatrix} 0 & \eta^b \end{pmatrix} \begin{pmatrix} & \delta^{\dot{b}a} \\ \delta_b^a & \end{pmatrix} = \begin{pmatrix} \eta^a & 0 \end{pmatrix} = \overline{(\psi^C)_R}$$

$$\overline{(\psi_R)^C} = [(\psi_R)^C]^\dagger\gamma^0 = \begin{pmatrix} \zeta_b^\dagger & 0 \end{pmatrix} \begin{pmatrix} & \delta^{\dot{b}a} \\ \delta_b^a & \end{pmatrix} = \begin{pmatrix} 0 & \zeta_a^\dagger \end{pmatrix} = \overline{(\psi^C)_L}$$

构造 Majorana 质量项

 由此推出

$$\begin{aligned}
\overline{(\psi_L)^C}(\psi_L)^C &= \begin{pmatrix} \eta^a & 0 \end{pmatrix} \begin{pmatrix} 0 \\ \eta^{\dagger \dot{a}} \end{pmatrix} = 0, & \overline{(\psi_R)^C}(\psi_R)^C &= \begin{pmatrix} 0 & \zeta_a^\dagger \end{pmatrix} \begin{pmatrix} \zeta_a \\ 0 \end{pmatrix} = 0 \\
\overline{(\psi_R)^C}\psi_L &= \begin{pmatrix} 0 & \zeta_a^\dagger \end{pmatrix} \begin{pmatrix} \eta_a \\ 0 \end{pmatrix} = 0, & \overline{(\psi_L)^C}\psi_R &= \begin{pmatrix} \eta^a & 0 \end{pmatrix} \begin{pmatrix} 0 \\ \zeta^{\dagger \dot{a}} \end{pmatrix} = 0 \\
\overline{\psi_R}(\psi_L)^C &= \begin{pmatrix} \zeta^a & 0 \end{pmatrix} \begin{pmatrix} 0 \\ \eta^{\dagger \dot{a}} \end{pmatrix} = 0, & \overline{\psi_L}(\psi_R)^C &= \begin{pmatrix} 0 & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} \zeta_a \\ 0 \end{pmatrix} = 0
\end{aligned}$$

 以上 6 个算符**不能**用于构造质量项。可用的 **Majorana 质量项算符**是

$$\begin{aligned}
\overline{(\psi_L)^C}\psi_L &= \begin{pmatrix} \eta^a & 0 \end{pmatrix} \begin{pmatrix} \eta_a \\ 0 \end{pmatrix} = \eta\eta, & \overline{\psi_L}(\psi_L)^C &= \begin{pmatrix} 0 & \eta_a^\dagger \end{pmatrix} \begin{pmatrix} 0 \\ \eta^{\dagger \dot{a}} \end{pmatrix} = \eta^\dagger\eta^\dagger \\
\overline{(\psi_R)^C}\psi_R &= \begin{pmatrix} 0 & \zeta_a^\dagger \end{pmatrix} \begin{pmatrix} 0 \\ \zeta^{\dagger \dot{a}} \end{pmatrix} = \zeta^\dagger\zeta^\dagger, & \overline{\psi_R}(\psi_R)^C &= \begin{pmatrix} \zeta^a & 0 \end{pmatrix} \begin{pmatrix} \zeta_a \\ 0 \end{pmatrix} = \zeta\zeta
\end{aligned}$$

 $\overline{(\psi_L)^C}\psi_L$ 与 $\overline{\psi_L}(\psi_L)^C$ 互为厄米共轭，而 $\overline{(\psi_R)^C}\psi_R$ 与 $\overline{\psi_R}(\psi_R)^C$ 互为厄米共轭

组合成 Majorana 旋量场

 将自由 Majorana 旋量场的拉氏量改写为

$$\mathcal{L}_{M1} = i\eta^\dagger \bar{\sigma}^\mu \partial_\mu \eta - \frac{m}{2}(\eta\eta + \eta^\dagger \eta^\dagger) = i\bar{\psi}_L \gamma^\mu \partial_\mu \psi_L - \frac{m}{2} \left[\overline{(\psi_L)^C} \psi_L + \overline{\psi_L} (\psi_L)^C \right]$$

 它描述 Majorana 旋量场 $\psi_1 \equiv \psi_L + (\psi_L)^C = \begin{pmatrix} \eta_a \\ \eta^{\dagger \dot{a}} \end{pmatrix}$ 的自由运动

 另一方面, 描述 Majorana 旋量场 $\psi_2 \equiv \psi_R + (\psi_R)^C = \begin{pmatrix} \zeta_a \\ \zeta^{\dagger \dot{a}} \end{pmatrix}$ 自由运动的拉氏量是

$$\mathcal{L}_{M2} = i\zeta \sigma^\mu \partial_\mu \zeta^\dagger - \frac{m}{2}(\zeta\zeta + \zeta^\dagger \zeta^\dagger) = i\bar{\psi}_R \gamma^\mu \partial_\mu \psi_R - \frac{m}{2} \left[\overline{\psi_R} (\psi_R)^C + \overline{(\psi_R)^C} \psi_R \right]$$